

Full-production discretization P3: Hessian/Ritz and hardware precision

TECT verification-first repository · claim A3-FULL-PRODUCTION-DISCRETIZATION-CONTINUUM

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1. Purpose and scope

This supplement records P3 stages 3 and 4. The domain, functional, positive floors, eta-shell-zero rule, and real pairing are unchanged from v1.0. Stage 3 asks which conditions make a finite-grid Ritz value interpretable as a Hessian spectral value. Stage 4 compares an independent device/dtype-variable implementation with the canonical CPU complex128 backend.

1 Hessian/Ritz conditions and result

At a homogeneous nonzero Class-II-active background, translation invariance makes the cosine/sine block for Fourier index $(1, 1, 0)$ invariant when all six internal real coordinates are retained. The real block dimension is 12. If a Ritz cluster has gap $g > 0$ and residual norm $r < g/2$, the standard residual bound localises true spectrum within distance r and keeps the cluster separated from its complement. The audit measures both quantities rather than inferring eigenvalue validity from a symmetric small matrix.

Across N8, N12, N16 and N20, the block has maximum invariance defect $5.45 \cdot 10^{-8}$ and relative anti-symmetry below $7.50 \cdot 10^{-11}$. The lowest cluster has multiplicity two, gap $g = 0.1368657$, maximum residual $1.377 \cdot 10^{-9}$, and $r/g = 1.0061 \cdot 10^{-8}$. A separate nonuniform stress field with Fourier indices through three exposes aliasing at N8. Its fixed eight-real dimensional Ritz errors against N20 are $1.5199 \cdot 10^{-3}$, $4.7888 \cdot 10^{-7}$, and $2.7136 \cdot 10^{-8}$, with observed orders 19.89 and 9.98. The nonuniform subspace is not invariant and its values are not labelled as full-spectrum eigenvalues.

2 Hardware and precision result

The portable implementation reproduces the functional independently of the P1 backend and makes device and complex dtype explicit. On homogeneous and nonuniform Class-II-active N8 fields, CPU complex128 agrees exactly with the canonical energy and residual. CPU complex64 has maximum relative energy error $9.8340 \cdot 10^{-8}$ and residual error $2.8775 \cdot 10^{-7}$, below the frozen $5 \cdot 10^{-6}$ and $5 \cdot 10^{-5}$ limits.

The installed runtime is Torch 2.11.0+cpu, with no CUDA device exposed. CUDA complex128 and complex64 are therefore recorded as UNAVAILABLE, not PASS. The same executable automatically runs those rows when a CUDA-enabled Torch build and device are present.

3 Devil’s-advocate and mandatory adversarial review

Sign and factor. The Hessian action is the centred derivative of the canonical real gradient, and the Ritz matrix uses the same cell-volume real pairing. Antisymmetry is measured before symmetrisation. **Convention.** All six internal real directions and both cosine/sine profiles are retained; dropping either would create a false invariant block. **Units.** Ritz values use the pinned dimensionless fixed-torus normalization. **Convergence.** The first stress field placed all grids at the finite-difference floor and was rejected

despite 11/13 passing; a higher-bandwidth declared test input exposes measurable convergence without relaxing thresholds. **Hardcode masking.** No measured eigenvalue, gap, residual, error, or rate is an input. Modes and thresholds are declared test inputs in the manifest. **Limit cases.** This does not cover a BCC candidate Hessian, a non-isolated cluster, spectral pollution for arbitrary expanding subspaces, rho-floor removal, or unavailable CUDA hardware. External review of the invariant-block argument and the portable complex-gradient convention is invited.

4 Reproduction

Run the Hessian/Ritz script followed by the hardware/precision script. Expected local verdicts are 13/13 Hessian/Ritz PASS and CPU-PRECISION-PASS-GPU-UNAVAILABLE.

Result footer

Result ID:	A3-FULL-PRODUCTION-DISCRETIZATION-CONTINUUM
Precise statement:	Stage 3 Hessian/Ritz passes in the declared invariant and fixed-subspace scopes; CPU precision rows pass.
Scope:	Fixed T3, pinned full-production functional and floors, eta_shell=0, N8/12/16 with N20 Ritz reference.
Dependencies:	P1 canonical backend and P2 continuum PDE package.
Evidence grade:	ANALYTIC + EXECUTED + CONDITIONAL; T3 scaffold.
Reproduction command:	python codes/foundations/ a3_full_production_hessian_ritz_convergence.py python codes/foundations/ a3_full_production_hardware_precision.py
Expected output:	13/13 PASS; CPU rows PASS; CUDA PASS or UNAVAILABLE.
Falsification gate:	A3-FULL-DISCRETIZATION-CLOSURE.
Tier before / after:	T3 / T3.
No-overclaim statement:	Not full candidate spectrum, GPU equivalence, independent continuum residual, T5, T6, or T7.
Next required action:	Execute CUDA rows, then independent continuum residual and integrated convergence proof.